

Course Overview

ITSC 3155 – Software Engineering
Department of Computer Science
College of Computing and Informatics



About Teaching Team

What Will You Learn?

Lectures

- Software Engineering Principles
- Software Development Life Cycle
- Software Requirements
- Software Design Basics and Approaches
- Agile Software Development & Scrum
- Software Testing
- Software implementation and deployment
- Software Security
- Containerization
- DevOps
- Cloud Computing
- Machine Learning

Professional Software Development

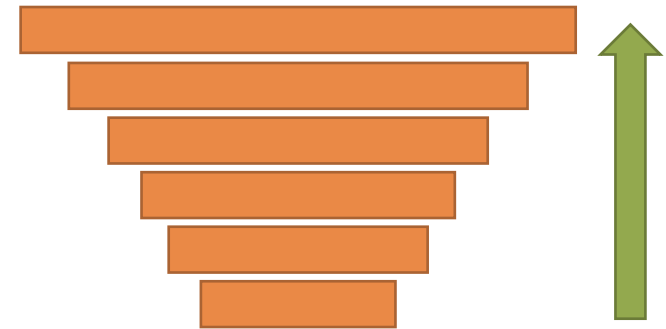
Engineer Your Software!

What Will You Learn?

Recitation

- Object Oriented Programming (OOP)
- Version Control
- Web Development
- APIs
- Database
- Full Stack Development
- Test Driven Development (TDD)
- Docker

All of concepts



Professional Software Development
Engineer Your Software!



Course Infrastructure and logistics

Infrastructure:

- Canvas: schedule, slides, syllabus, office hours, homework, grades, other materials.
- Git/Github for coding and collaboration

Logistics:

- Participate in Lecture
- Participate in Recitations
- Office hours every weekday!

Best way of communication is email. CC all the TAs.

Prerequisites & Programming Skills

Basic knowledge of programming language



This is not a programming class! BUT don't worry!
Follow the course materials!

Tools



PyCharm



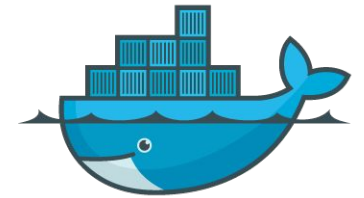
Workbench



WebStorm



DataGrip



docker

Docker Desktop

Optional (Highly
Recommended)

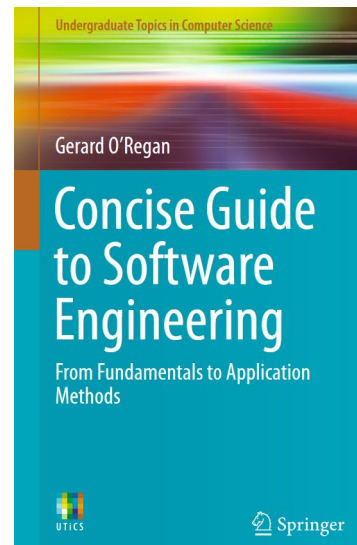
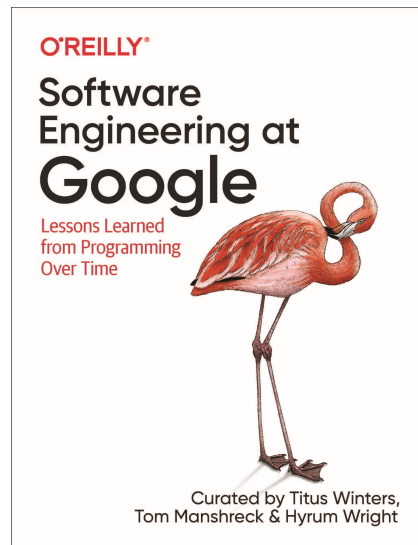
Activate your student account on [JetBrains](#) website and download ultimate versions for free!

Limited versions cost around \$300 annually!

Textbook

NOT mandatory

- Winters, Titus, Tom Manshreck, and Hyrum Wright. [Software Engineering at Google](#): Lessons learned from programming over time. O'Reilly Media, 2020.
- O'Regan, Gerard. [Concise guide to software engineering](#). Springer, Cham, 2017.



Modules Structure

Prep Work

Preparation for the module (installing tools, guidelines, surveys)

Lecture

Conceptual, Theoretical, Best Practice

Recitation

Practical tasks, preparation for homework, discussion.

Homework

Optional programming videos, Assignments, Quizzes.

Attendance

You need to submit your attendance during the class.

Additional Resources

Cheat sheets, Articles, New technologies

Bonus

Job interview prep, Extra credit assignments

Assignments

Individual Assignments

- Coding and implementation
- Develop design document
- Develop plans

Group projects

- Implementing a full-stack software project from scratch! (starts second half of semester)
- Standard Collaboration Policy
 - In group work, be honest about contribution of group members; do not cover for others

Professionalism

- Being a professional means you should work well with others.
- The best professionals are those who make those around them better.
- If you feel someone is not treating you or someone else in a professional manner, you have two options:
 - If you feel you have the standing to do so, speak up!
 - Reach out to us, and we will meet with you privately to discuss it, as well as preserve your anonymity.

Late Day Policy (Assignments only)

- You will lose 5% of grade for each day.
- One week after the due date is the deadline for late submissions (30% penalty). You will then receive a 0.
- Exemption in case of health issues, travel for interviews, ... on case by case **if you inform us at least 2 day before deadline.**

There is no late submission allowed for quizzes!

Grading

- **Homework – 30%**
- **Quizzes – 20%**
- **Midterm – 15%**
- **Final Project – 30%**
- **Attendance – 5%**

Support System

- TA Hours – Everyday
- CCI Tutoring Center
- Peers
- The Internet , **Stack Overflow**, **AI Chatbots**
- Email me to book a one-on-one appointment.

ChatGPT/Copilot/etc Policy

- You are NOT allowed to use ChatGPT (or Copilot, etc.) for the quizzes.
- You are NOT allowed to use ChatGPT for generating code from scratch.
- For the coding assignment, you are allowed to use ChatGPT to debug your code.

